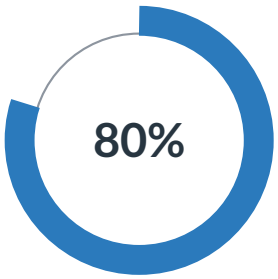
Results

Carter Feng — 1st Attempt



80
Out of 100 points

01:06:47
Time for this attempt

This assessment is locked.

Attempt History

Results	Points	Score	(First score is kept)
Attempt 1	80 of 100	80%	(First score)

Your Answers:

1 3.333 / 10 points Matching

SET 2: 929 873 996 200 846 441 21 481 789 91 729 961 786 41 36 147

Create a red-black tree using the following values: 929 873 996 200

What are the following node values for the 7th color swap?

Black X Node	✗ 91	Correct Answer: **200**
Red Left Child Node	✓ 41	
Red Right Child Node	✗ 147	Correct Answer: **481**

2 10 / 10 points Matching

SET 2: 929 873 996 200 846 441 21 481 789 91 729 961 786 41 36 147

Create a red-black tree using the following values: 929 873 996 200

What are the first 10 nodes printed in a level order traversal?

1	2	3	4	5	6	7	8	9	10

1	✓ 846
2	✓ 200
3	✓ 929
4	✓ 41
5	✓ 481

6	✓	873
7	✓	996
8	✓	21
9	✓	143
10	✓	441

3 10 / 10 points Matching

SET 2: 929 873 996 200 846 441 21 481 789 91 729 961 786 41 36 147

Create a red-black tree using the following values: 929 873 996 200

What are the last 10 nodes printed in a level order traversal?

1	2	3	4	5	6	7	8	9	10

1	✓	786
2	✓	961
3	✓	36

4	✓ 91
5	✓ 147
6	✓ 207
7	✓ 729
8	✓ 789
9	✓ 65
10	✓ 530

4 10 / 10 points Matching

SET 2: 929 873 996 200 846 441 21 481 789 91 729 961 786 41 36 147

Create a red-black tree using the following values: 929 873 996 200

What are the first 10 nodes printed in a post order traversal?

1	2	3	4	5	6	7	8	9	10

1	✓ 36
2	✓ 21

3	✓	65
4	✓	91
5	✓	147
6	✓	143
7	✓	41
8	✓	207
9	✓	441
10	✓	530

5 6.667 / 10 points Matching

SET 2: 929 873 996 200 846 441 21 481 789 91 729 961 786 41 36 147

Create a red-black tree using the following values: 929 873 996 200

What are the following node values for the 3rd color swap?

Black X Node	✓	200

Red Left Child Node	✗ null	Correct Answer: **21**
Red Right Child Node	✓ 441	

6 10 / 10 points Fill in the Blank

SET 2: 929 873 996 200 846 441 21 481 789 91 729 961 786 41 36 147

Create a red-black tree using the following values: 929 873 996 200

What is the height of the tree?

✓ 5

7 0 / 10 points Fill in the Blank

SET 2: 929 873 996 200 846 441 21 481 789 91 729 961 786 41 36 147

Create a red-black tree using the following values: 929 873 996 200

How many red nodes of degree 0 are in the tree?

✗ 8

Correct Answer: **5**

8 10 / 10 points Matching

SET 2: 929 873 996 200 846 441 21 481 789 91 729 961 786 41 36 147

Create a red-black tree using the following values: 929 873 996 200

What are the nodes from left to right on level 1 of the red black t

1	2

1	✓ 200
2	✓ 929

9 10 / 10 points Fill in the Blank

SET 2: 929 873 996 200 846 441 21 481 789 91 729 961 786 41 36 147

Create a red-black tree using the following values: 929 873 996 200

What is the diameter of the tree?

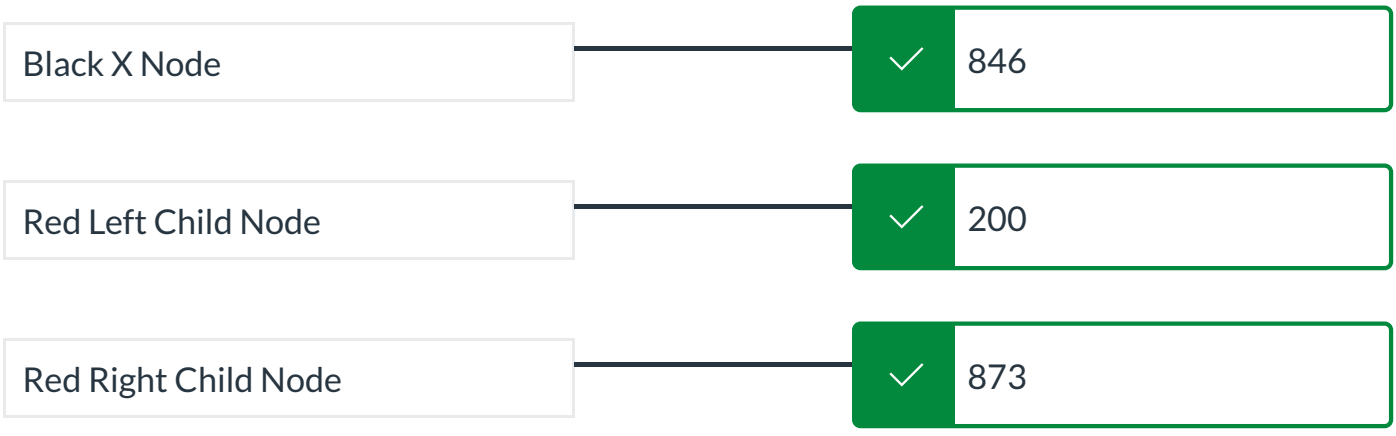
✓ 9

10 10 / 10 points Matching

SET 2: 929 873 996 200 846 441 21 481 789 91 729 961 786 41 36 147

Create a red-black tree using the following values: 929 873 996 200

What are the following node values for the 2nd color swap?


Black X Node
846
Red Left Child Node
200
Red Right Child Node
873